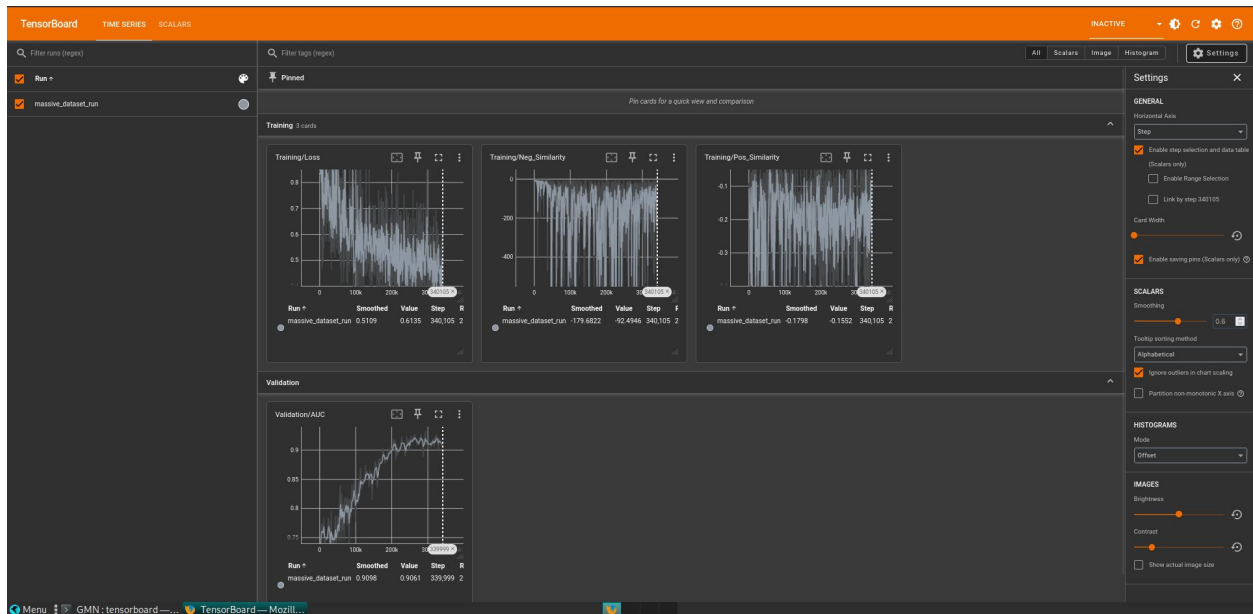
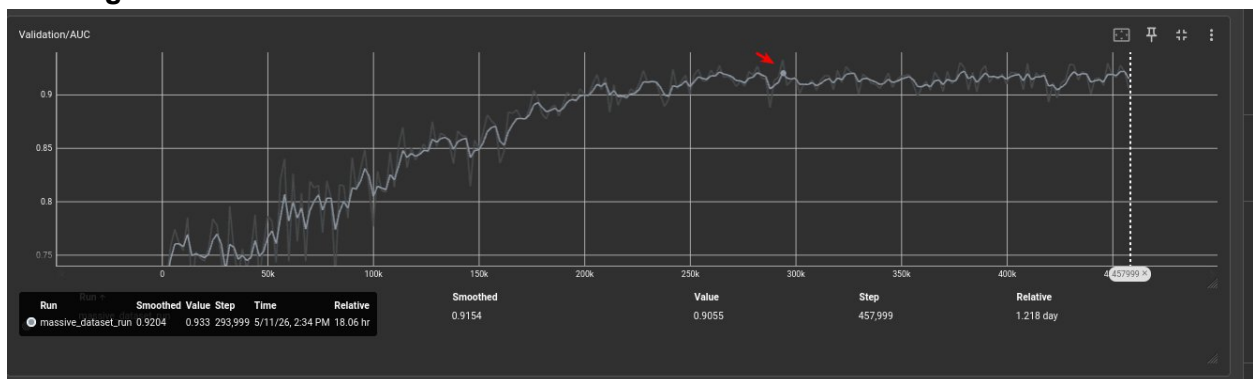


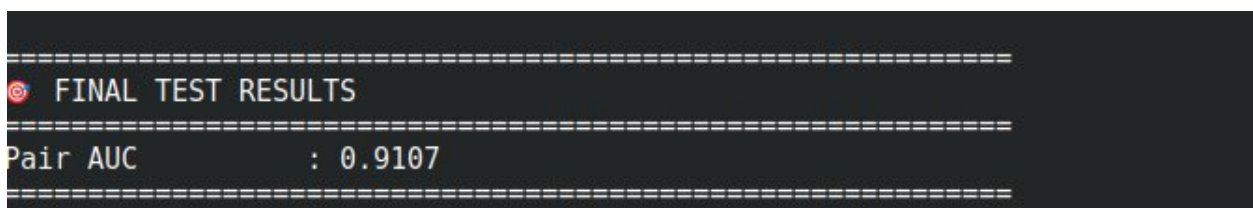
Tensorboard



Training best AUC:



AUC on test set



It didn't memorize just the training data (overfitting); it generalized beautifully to unseen data.

Testing with big functions

Examples:

```
curl_curl_easy_setopt (gcc_01)
```

```
wget_main (clang_02)
```

```
(.env) [kh4n@parrot]~/research/Graph-Matching-Networks/GMN
└─$ python infer.py
[*] Using device: cuda:0
[*] Loading best model...
✓ Best model loaded successfully!

[*] Scanning database compiled binaries/test database.sqlite...
[*] Found 1346 unique functions capable of forming pairs.

Available functions (first 10):
['curl_GetFileAndPassword', 'curl_GetSizeParameter', 'curl_Memdup', 'curl_SetHTTPrequest', 'curl_add2list', 'curl_add_file_name_to_url', 'curl_add_parallel_transfers', 'curl_add_per_transfer', 'curl_addvariable', 'curl_base64_encode', 'curl_bsearch', 'curl_c_escape', 'curl_call_oc', 'curl_check_protocol', 'curl_checkhome', 'curl_checkpasswd', 'curl_clean_getout', 'curl_cleanarg', 'curl_clock_gettime', 'curl_close', 'curl_config_free', 'curl_config_init', 'curl_create_dir_hierarchy', 'curl_create_transfer', 'curl_curl_easy_cleanup', 'curl_curl_easy_escape', 'curl_curl_easy_getinfo', 'curl_curl_easy_header', 'curl_curl_easy_init', 'curl_curl_easy_nextheader', 'curl_curl_easy_pause', 'curl_curl_easy_perform', 'curl_curl_easy_setopt', 'curl_curl_easy_strerror', 'curl_curl_free', 'curl_curl_getdate', 'curl_curl_getenv', 'curl_curl_global_cleanup', 'curl_curl_global_init', 'curl_curl_global_trace', 'curl_curl_maprintf', 'curl_curl_mfprintf', 'curl_curl_mime_addpart', 'curl_curl_mime_data', 'curl_curl_mime_data_cb', 'curl_curl_mime_encoder', 'curl_curl_mime_filedata', 'curl_curl_mime_filename', 'curl_curl_mime_free', 'curl_curl_mime_headers']

=====
🔍 COMPARING: 'curl_curl_easy_setopt' vs 'wget_main'
=====

🔴 AI Similarity Score: -107.5050
  (Higher score = More mathematically similar)
=====
```

```
curl_curl_easy_setopt gcc_01
```

```
curl_curl_easy_setopt clang_02
```

```
(.env) [kh4n@parrot]~/research/Graph-Matching-Networks/GMN
└─$ python infer.py
[*] Using device: cuda:0
[*] Loading best model...
✓ Best model loaded successfully!

[*] Scanning database compiled binaries/test database.sqlite...
[*] Found 1346 unique functions capable of forming pairs.

Available functions (first 10):
['curl_GetFileAndPassword', 'curl_GetSizeParameter', 'curl_Memdup', 'curl_SetHTTPrequest', 'curl_add2list', 'curl_add_file_name_to_url', 'curl_add_parallel_transfers', 'curl_add_per_transfer', 'curl_addvariable', 'curl_base64_encode', 'curl_bsearch', 'curl_c_escape', 'curl_call_oc', 'curl_check_protocol', 'curl_checkhome', 'curl_checkpasswd', 'curl_clean_getout', 'curl_cleanarg', 'curl_clock_gettime', 'curl_close', 'curl_config_free', 'curl_config_init', 'curl_create_dir_hierarchy', 'curl_create_transfer', 'curl_curl_easy_cleanup', 'curl_curl_easy_escape', 'curl_curl_easy_getinfo', 'curl_curl_easy_header', 'curl_curl_easy_init', 'curl_curl_easy_nextheader', 'curl_curl_easy_pause', 'curl_curl_easy_perform', 'curl_curl_easy_setopt', 'curl_curl_easy_strerror', 'curl_curl_free', 'curl_curl_getdate', 'curl_curl_getenv', 'curl_curl_global_cleanup', 'curl_curl_global_init', 'curl_curl_global_trace', 'curl_curl_maprintf', 'curl_curl_mfprintf', 'curl_curl_mime_addpart', 'curl_curl_mime_data', 'curl_curl_mime_data_cb', 'curl_curl_mime_encoder', 'curl_curl_mime_filedata', 'curl_curl_mime_filename', 'curl_curl_mime_free', 'curl_curl_mime_headers']

=====
🔍 COMPARING: 'curl_curl_easy_setopt' vs 'wget_main'
=====

🔴 AI Similarity Score: -0.0000
  (Higher score = More mathematically similar)
=====
```

```
wget_main gcc_01
```

```
wget_main clang_02
```

```
(.env) [kh4n@parrot]~/research/Graph-Matching-Networks/GMN]
└─$ python infer.py
[*] Using device: cuda:0
[*] Loading best model...
✓ Best model loaded successfully!

[*] Scanning database compiled binaries/test_database.sqlite...
[*] Found 1346 unique functions capable of forming pairs.

Available functions (first 10):
['curl_GetFileAndPassword', 'curl_GetSizeParameter', 'curl_Memdup', 'curl_SetHTTPrequest', 'curl_add2list', 'curl_add_file_name_to_url', 'c
url_add_parallel_transfers', 'curl_add_per_transfer', 'curl_addvariable', 'curl_base64_encode', 'curl_bsearch', 'curl_c_escape', 'curl_call
oc', 'curl_check_protocol', 'curl_checkhome', 'curl_checkpasswd', 'curl_clean_getout', 'curl_cleanarg', 'curl_clock_gettime', 'curl_close',
'curl_config_free', 'curl_config_init', 'curl_create_dir_hierarchy', 'curl_create_transfer', 'curl_curl_easy_cleanup', 'curl_curl_easy_esc
ape', 'curl_curl_easy_getinfo', 'curl_curl_easy_header', 'curl_curl_easy_init', 'curl_curl_easy_nextheader', 'curl_curl_easy_pause', 'curl
curl_easy_perform', 'curl_curl_easy_setopt', 'curl_curl_easy_strerror', 'curl_curl_free', 'curl_curl_getdate', 'curl_curl_getenv', 'curl_cu
rl_global_cleanup', 'curl_curl_global_init', 'curl_curl_global_trace', 'curl_curl_maprintf', 'curl_curl_mfprintf', 'curl_curl_mime_addpart'
, 'curl_curl_mime_data', 'curl_curl_mime_data_cb', 'curl_curl_mime_encoder', 'curl_curl_mime_filedata', 'curl_curl_mime_filename', 'curl_cu
rl_mime_free', 'curl_curl_mime_headers']

=====
❏ COMPARING: 'curl_curl_easy_setopt' vs 'wget_main'
=====

● AI Similarity Score: -0.0298
  (Higher score = More mathematically similar)
=====
```

curl_main gcc01

Wget_http clang_02

```
(.env) [kh4n@parrot]~/research/Graph-Matching-Networks/GMN]
└─$ python infer.py
[*] Using device: cuda:0
[*] Loading best model...
✓ Best model loaded successfully!

[*] Scanning database compiled binaries/test_database.sqlite...
[*] Found 1346 unique functions capable of forming pairs.

Available functions (first 10):
['curl_GetFileAndPassword', 'curl_GetSizeParameter', 'curl_Memdup', 'curl_SetHTTPrequest', 'curl_add2list', 'curl_add_file_name_to_url', 'c
url_add_parallel_transfers', 'curl_add_per_transfer', 'curl_addvariable', 'curl_base64_encode', 'curl_bsearch', 'curl_c_escape', 'curl_call
oc', 'curl_check_protocol', 'curl_checkhome', 'curl_checkpasswd', 'curl_clean_getout', 'curl_cleanarg', 'curl_clock_gettime', 'curl_close',
'curl_config_free', 'curl_config_init', 'curl_create_dir_hierarchy', 'curl_create_transfer', 'curl_curl_easy_cleanup', 'curl_curl_easy_esc
ape', 'curl_curl_easy_getinfo', 'curl_curl_easy_header', 'curl_curl_easy_init', 'curl_curl_easy_nextheader', 'curl_curl_easy_pause', 'curl
curl_easy_perform', 'curl_curl_easy_setopt', 'curl_curl_easy_strerror', 'curl_curl_free', 'curl_curl_getdate', 'curl_curl_getenv', 'curl cu
rl_global_cleanup', 'curl_curl_global_init', 'curl_curl_global_trace', 'curl_curl_maprintf', 'curl_curl_mfprintf', 'curl_curl_mime_addpart'
, 'curl_curl_mime_data', 'curl_curl_mime_data_cb', 'curl_curl_mime_encoder', 'curl_curl_mime_filedata', 'curl_curl_mime_filename', 'curl cu
rl_mime_free', 'curl_curl_mime_headers']

=====
❏ COMPARING: 'curl_main' vs 'wget_gethttp'
=====

● AI Similarity Score: -4.6989
  (Higher score = More mathematically similar)
=====
```

curl_single_transfer gcc_01

wget_ftp_retrieve_glob clang_02


```
(.env) [kh4n@parrot]~/research/Graph-Matching-Networks/GMN
└─$ python infer.py
[*] Using device: cuda:0
[*] Loading best model...
✓ Best model loaded successfully!

[*] Scanning database compiled binaries/test database.sqlite...
[*] Found 1346 unique functions capable of forming pairs.

Available functions (first 10):
['curl_GetFileAndPassword', 'curl_GetSizeParameter', 'curl_Memdup', 'curl_SetHTTPrequest', 'curl_add2list', 'curl_add_file_name_to_url', 'c
url_add_parallel_transfers', 'curl_add_per_transfer', 'curl_addvariable', 'curl_base64_encode', 'curl_bsearch', 'curl_c_escape', 'curl_call
oc', 'curl_check_protocol', 'curl_checkhome', 'curl_checkpasswd', 'curl_clean_getout', 'curl_cleanarg', 'curl_clock_gettime', 'curl_close',
'curl_config_free', 'curl_config_init', 'curl_create_dir_hierarchy', 'curl_create_transfer', 'curl_curl_easy_cleanup', 'curl_curl_easy_esc
ape', 'curl_curl_easy_getinfo', 'curl_curl_easy_header', 'curl_curl_easy_init', 'curl_curl_easy_nextheader', 'curl_curl_easy_pause', 'curl
curl_easy_perform', 'curl_curl_easy_setopt', 'curl_curl_easy_strerror', 'curl_curl_free', 'curl_curl_getdate', 'curl_curl_getenv', 'curl_cu
rl_global_cleanup', 'curl_curl_global_init', 'curl_curl_global_trace', 'curl_curl_maprintf', 'curl_curl_mfprintf', 'curl_curl_mime_addpart'
, 'curl_curl_mime_data', 'curl_curl_mime_data_cb', 'curl_curl_mime_encoder', 'curl_curl_mime_filedata', 'curl_curl_mime_filename', 'curl_cu
rl_mime_free', 'curl_curl_mime_headers']

=====
🔍 COMPARING: 'curl_single_transfer' vs 'wget_ftp_retrieve_glob'
=====

🔴 AI Similarity Score: -2.5254
(Higher score = More mathematically similar)
=====
```

wget_ftp_retrieve_glob gcc_01

wget_ftp_retrieve_glob clang_02

```
(.env) [kh4n@parrot]~/research/Graph-Matching-Networks/GMN
└─$ python infer.py
[*] Using device: cuda:0
[*] Loading best model...
✓ Best model loaded successfully!

[*] Scanning database compiled binaries/test database.sqlite...
[*] Found 1346 unique functions capable of forming pairs.

Available functions (first 10):
['curl_GetFileAndPassword', 'curl_GetSizeParameter', 'curl_Memdup', 'curl_SetHTTPrequest', 'curl_add2list', 'curl_add_file_name_to_url', 'c
url_add_parallel_transfers', 'curl_add_per_transfer', 'curl_addvariable', 'curl_base64_encode', 'curl_bsearch', 'curl_c_escape', 'curl_call
oc', 'curl_check_protocol', 'curl_checkhome', 'curl_checkpasswd', 'curl_clean_getout', 'curl_cleanarg', 'curl_clock_gettime', 'curl_close',
'curl_config_free', 'curl_config_init', 'curl_create_dir_hierarchy', 'curl_create_transfer', 'curl_curl_easy_cleanup', 'curl_curl_easy_esc
ape', 'curl_curl_easy_getinfo', 'curl_curl_easy_header', 'curl_curl_easy_init', 'curl_curl_easy_nextheader', 'curl_curl_easy_pause', 'curl
curl_easy_perform', 'curl_curl_easy_setopt', 'curl_curl_easy_strerror', 'curl_curl_free', 'curl_curl_getdate', 'curl_curl_getenv', 'curl cu
rl_global_cleanup', 'curl_curl_global_init', 'curl_curl_global_trace', 'curl_curl_maprintf', 'curl_curl_mfprintf', 'curl_curl_mime_addpart'
, 'curl_curl_mime_data', 'curl_curl_mime_data_cb', 'curl_curl_mime_encoder', 'curl_curl_mime_filedata', 'curl_curl_mime_filename', 'curl_cu
rl_mime_free', 'curl_curl_mime_headers']

=====
🔍 COMPARING: 'curl_single_transfer' vs 'wget_ftp_retrieve_glob'
=====

🔴 AI Similarity Score: -0.0342
(Higher score = More mathematically similar)
=====
```

Stripped binaries

```
(.env) [kh4n@parrot]~/research/Graph-Matching-Networks/GMN
└─$ python infer.py
[*] Using device: cuda:0
[*] Loading best model...
✓ Best model loaded successfully!

[*] Scanning database compiled binaries/test database.sqlite...
[*] Found 1346 unique functions capable of forming pairs.

Available functions (first 50 curl functions):
['curl_GetFileAndPassword', 'curl_GetSizeParameter', 'curl_Memdup', 'curl_SetHTTPrequest', 'curl_add2list', 'curl_add_file_name_to_url', 'curl_add_parallel_transfers', 'curl_add_per_transfer', 'curl_addvariable', 'curl_base64_encode', 'curl_bsearch', 'curl_c_escape', 'curl_call_oc', 'curl_check_protocol', 'curl_checkhome', 'curl_checkpasswd', 'curl_clean_getout', 'curl_cleanarg', 'curl_clock_gettime', 'curl_close', 'curl_config_free', 'curl_config_init', 'curl_create_dir_hierarchy', 'curl_create_transfer', 'curl_easy_cleanup', 'curl_easy_escape', 'curl_easy_getinfo', 'curl_easy_header', 'curl_easy_init', 'curl_easy_nextheader', 'curl_easy_pause', 'curl_easy_perform', 'curl_easy_setopt', 'curl_easy_strerror', 'curl_free', 'curl_getdate', 'curl_getenv', 'curl_global_cleanup', 'curl_global_init', 'curl_global_trace', 'curl_maprintf', 'curl_mfprintf', 'curl_mime_addpart', 'curl_mime_data', 'curl_mime_data_cb', 'curl_mime_encoder', 'curl_mime_filedata', 'curl_mime_filename', 'curl_mime_free', 'curl_mime_headers']

Available functions (first 50 wget functions):
['wget_abort', 'wget_abort_run_with_timeout', 'wget_accdir', 'wget_accept', 'wget_accept_connection', 'wget_accept_domain', 'wget_accept_url', 'wget_acceptable', 'wget_add_path', 'wget_address_list_address_at', 'wget_address_list_connected_p', 'wget_address_list_contains', 'wget_address_list_contains_cold', 'wget_address_list_delete', 'wget_address_list_from_addrinfo', 'wget_address_list_get_bounds', 'wget_address_list_release', 'wget_address_list_set_connected', 'wget_address_list_set_faulty', 'wget_advance_declaration', 'wget_afalg_stream', 'wget_ajoin_dir_file', 'wget_alarm_cancel', 'wget_alarm_set', 'wget_append_char', 'wget_append_dir_structure', 'wget_append_null', 'wget_append_string', 'wget_append_uri_pathel', 'wget_append_uri_pathel_constprop_0', 'wget_append_url', 'wget_append_value_to_filename', 'wget_aprintf', 'wget_are_urls_equal', 'wget_atoi', 'wget_bar_create', 'wget_bar_draw', 'wget_bar_finish', 'wget_bar_set_params', 'wget_bar_update', 'wget_base32_decode_alloc_ctx', 'wget_base32_decode_ctx', 'wget_base32_decode_ctx_init', 'wget_base32_encode', 'wget_base32_encode_alloc', 'wget_base_len', 'wget_base_name', 'wget_basic_authentication_encode', 'wget_bcmp', 'wget_bind']

=====
❖ COMPARING: 'curl_cleanarg' vs 'wget_bar_set_params'
AI similarity match score range {0.0} --- {-1.2}
=====
✗ Error: One or both of the functions were not found in compiled binaries/test database.sqlite!
```

False Positives

curl_ipfs_url_rewrite gcc_01

Wget_url_parse clang_02

```
(.env) [kh4n@parrot]~/research/Graph-Matching-Networks/GMN
└─$ python infer.py
[*] Using device: cuda:0
[*] Loading best model...
✓ Best model loaded successfully!

[*] Scanning database compiled binaries/test database.sqlite...
[*] Found 1346 unique functions capable of forming pairs.

Available functions (first 10):
['curl_GetFileAndPassword', 'curl_GetSizeParameter', 'curl_Memdup', 'curl_SetHTTPrequest', 'curl_add2list', 'curl_add_file_name_to_url', 'curl_add_parallel_transfers', 'curl_add_per_transfer', 'curl_addvariable', 'curl_base64_encode', 'curl_bsearch', 'curl_c_escape', 'curl_call_oc', 'curl_check_protocol', 'curl_checkhome', 'curl_checkpasswd', 'curl_clean_getout', 'curl_cleanarg', 'curl_clock_gettime', 'curl_close', 'curl_config_free', 'curl_config_init', 'curl_create_dir_hierarchy', 'curl_create_transfer', 'curl_easy_cleanup', 'curl_easy_escape', 'curl_easy_getinfo', 'curl_easy_header', 'curl_easy_init', 'curl_easy_nextheader', 'curl_easy_pause', 'curl_easy_perform', 'curl_easy_setopt', 'curl_easy_strerror', 'curl_free', 'curl_getdate', 'curl_getenv', 'curl_global_cleanup', 'curl_global_init', 'curl_global_trace', 'curl_maprintf', 'curl_mfprintf', 'curl_mime_addpart', 'curl_mime_data', 'curl_mime_data_cb', 'curl_mime_encoder', 'curl_mime_filedata', 'curl_mime_filename', 'curl_mime_free', 'curl_mime_headers']

=====
❖ COMPARING: 'curl_ipfs_url_rewrite' vs 'wget_url_parse'
=====

❖ AI Similarity Score: -0.1418
(Higher score = More mathematically similar)
=====
```

curl_curl_mime_filedata gcc_01

wget_bind clang_02

```

(.env) [kh4n@parrot]~/research/Graph-Matching-Networks/GMN]
└─ $python infer.py
[*] Using device: cuda:0
[*] Loading best model...
✓ Best model loaded successfully!

[*] Scanning database compiled binaries/test database.sqlite...
[*] Found 1346 unique functions capable of forming pairs.

Available functions (first 50 curl functions):
['curl_GetFileAndPassword', 'curl_GetSizeParameter', 'curl_Memdup', 'curl_SetHTTPrequest', 'curl_add2list', 'curl_add_file_name_to_url', 'curl_add_parallel_transfers', 'curl_add_per_transfer', 'curl_addvariable', 'curl_base64_encode', 'curl_bsearch', 'curl_c_escape', 'curl_call_oc', 'curl_check_protocol', 'curl_checkhome', 'curl_checkpasswd', 'curl_clean_getout', 'curl_cleanarg', 'curl_clock_gettime', 'curl_close', 'curl_config_free', 'curl_config_init', 'curl_create_dir_hierarchy', 'curl_create_transfer', 'curl_easy_cleanup', 'curl_easy_escape', 'curl_easy_getinfo', 'curl_easy_header', 'curl_easy_init', 'curl_easy_nextheader', 'curl_easy_pause', 'curl_easy_perform', 'curl_easy_setopt', 'curl_easy_strerror', 'curl_free', 'curl_getdate', 'curl_getenv', 'curl_global_cleanup', 'curl_global_init', 'curl_global_trace', 'curl_maprintf', 'curl_mfprintf', 'curl_mime_addpart', 'curl_mime_data', 'curl_mime_data_cb', 'curl_mime_encoder', 'curl_mime_filedata', 'curl_mime_filename', 'curl_mime_free', 'curl_mime_headers']

Available functions (first 50 wget functions):
['wget_abort', 'wget_abort_run_with_timeout', 'wget_accdir', 'wget_accept', 'wget_accept_connection', 'wget_accept_domain', 'wget_accept_url', 'wget_acceptable', 'wget_add_path', 'wget_address_list_address_at', 'wget_address_list_connected_p', 'wget_address_list_contains', 'wget_address_list_contains_cold', 'wget_address_list_delete', 'wget_address_list_from_addrinfo', 'wget_address_list_get_bounds', 'wget_address_list_release', 'wget_address_list_set_connected', 'wget_address_list_set_faulty', 'wget_advance_declaration', 'wget_afile_stream', 'wget_ajoin_dir_file', 'wget_alarm_cancel', 'wget_alarm_set', 'wget_append_char', 'wget_append_dir_structure', 'wget_append_null', 'wget_append_string', 'wget_append_uri_pathel', 'wget_append_uri_pathel.constprop.0', 'wget_append_url', 'wget_append_value_to_filename', 'wget_aprintf', 'wget_are_urls_equal', 'wget_atoi', 'wget_bar_create', 'wget_bar_draw', 'wget_bar_finish', 'wget_bar_set_params', 'wget_bar_update', 'wget_base32_decode_alloc_ctx', 'wget_base32_decode_ctx', 'wget_base32_decode_ctx_init', 'wget_base32_encode', 'wget_base32_encode_alloc', 'wget_base_len', 'wget_base_name', 'wget_basic_authentication_encode', 'wget_bcmp', 'wget_bind']

=====
• COMPARING: 'curl_mime_filedata' vs 'wget_bind'
=====

• AI Similarity Score: -0.0000
  (Higher score = More mathematically similar)
=====

```